



Figure 6: Performance on different subsets. This figure indicates that curriculum learning is more beneficial to difficult data (larger k) than simple data (smaller k).

(MCD1/MCD2/MCD3) and all hyperparameter settings ($c=2000/2500/3000/3500/4000$). For MCD1/MCD2/MCD3 tasks of CFQ, CIBT outperforms IBT by 1.8%/11.3%/5.8% accuracy scores. An interesting observation is that incorporating curriculum learning into iterative back-translation brings more stable performance: IBT performs much worse on MCD2 (57.8%) than on MCD1/MCD3 (64.8%/64.6%), while CIBT’s improvement on MCD2 (11.3%) is much better than on MCD1/MCD3 (1.8%/5.8%), making the performances on MCD1/MCD2/MCD3 comparable (66.6%/69.1%/70.4%).

Curriculum learning is more beneficial to difficult data than simple data. Figure 6 further illustrates how curriculum learning improves the performance. We use k to denote test cases that would be put into the first k curriculums, then obtain 6 subsets of test data: $T_1, T_2, \dots, T_6$. We have $T_1 \subset T_2 \subset \dots \subset T_6$. As we can observe in Figure 6, on the simplest subset (T_1), the performances of IBT and CIBT are very close. If we consider T_2 , a subset which is a bit more difficult than T_1 , CIBT performs slightly better than IBT. Similarly, we observe that: more difficult a subset of test data is, more gains CIBT brings. Therefore, we can conclude that curriculum learning is more beneficial to difficult data than simple data.

6 Related Work

6.1 Compositional Generalization

Recently, exploring compositional generalization of DNN models has attracted a large attention on different topics, e.g., visual-question answering (?), algebraic compositionality (?), and logic inference (?). In this work, we focus on compositional generalization in language, which is benchmarked by SCAN (?) and CFQ (?) tasks.

While mainstream DNN models are proved to exhibit limited compositional generalization ability in language (????), there have been many efforts to address this limitation, including design specialized architectures (????), data augmentation (?), meta-learning (?) and pre-training models (?). Comparing to these previous work, this paper incorporates semi-supervised learning into compositional generalization, which is task-agnostic and model-agnostic, thus improving the performance on both SCAN and CFQ benchmarks.

6.2 Iterative Back-translation

The idea of iterative back-translation dates back at least to *back-translation*, which simply augments parallel training data using pseudo-parallel data generated from only target-side monolingual data. Back-translation has been proven effective in statistical machine translation (??) and neural machine translation (????). Iterative back-translation (????) is an extension of back-translation, in which forward and backward direction models generate pseudo-parallel data for each other and improve each other, bringing stronger empirical performance.

To our best knowledge, this paper is the first to study iterative back-translation from the perspective of compositional generalization. We demonstrate that iterative back-translation can help seq2seq models make better generalizations to much more combinations, thus broadening the understanding of iterative back-translation.

7 Conclusion

In this paper, we revisit iterative back-translation from the perspective of compositional generalization. We find that iterative back-translation is an effective method that exploits a large amount of monolingual data to enable seq2seq models better generalize to more combinations beyond limited parallel data. We also find that iterative back-translation has the ability to increasingly correct errors in pseudo-parallel data, thus contributing to its success. To encourage this behaviour, we propose curriculum iterative back-translation, which explicitly improves the quality of pseudo-parallel data, thus further improving the performance.

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